



Web Programming

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Objective

- ▶ History
- ▶ Internet Protocol
- ▶ Client-Server Model
- ▶ Where is Internet ?
- ▶ Domain Name System
- ▶ Uniform Resource Locators (URL)
- ▶ Hypertext Transfer Protocol (HTTP)



Why Web Technology ?

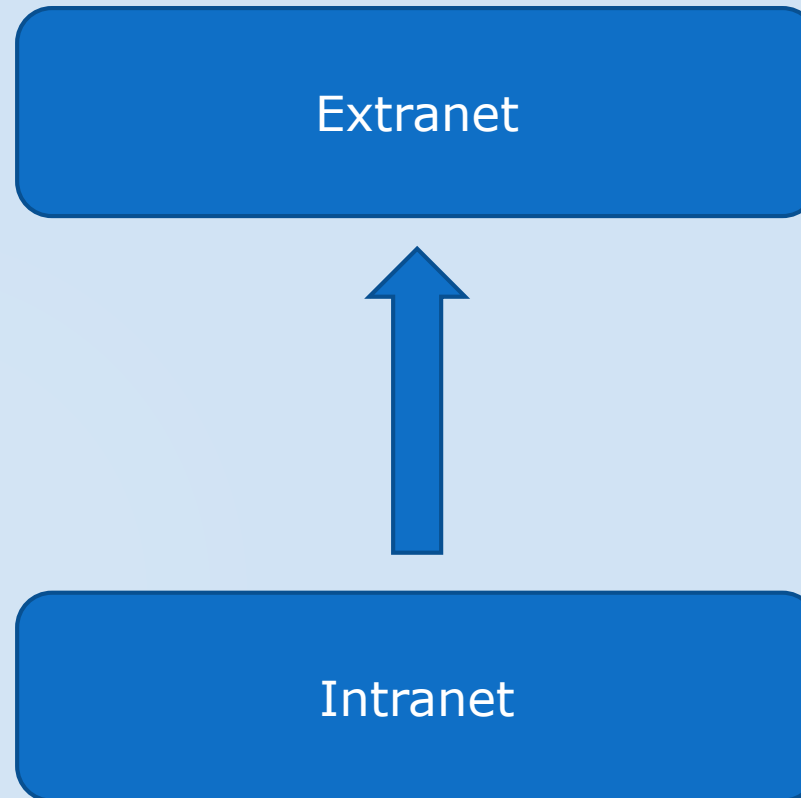
- ▶ In very short period of time, it has caused major changes in information technology, business processes, communications and the way many people spend their leisure time.
- ▶ To compare its growth, consider that the World Wide Web attained **50 millions users in four years.**
 - ▶ It took radio 38 years and television 13 years to reach similar penetration.



Processing Islands in the Enterprise

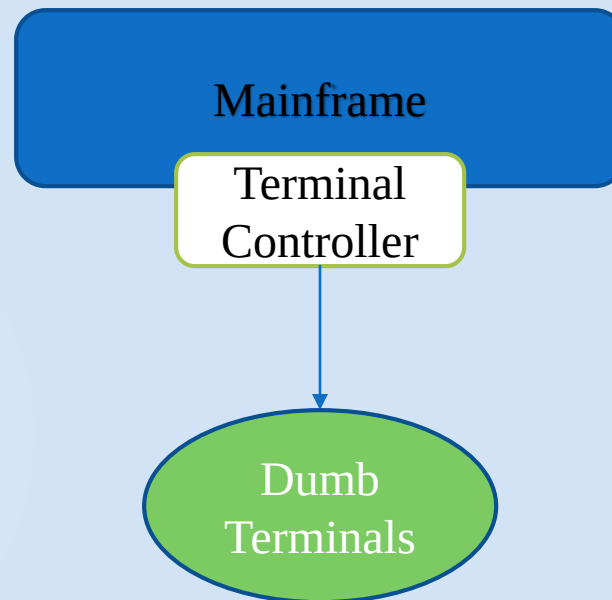
- ▶ **Data Processing** – the core IT systems that control the fundamental business processes in an organization.
- ▶ **Personal Productivity** – Employees works with information using various tools.
- ▶ **Collaboration** – the use of communications and GroupWare software that enables both organizations and individuals to work in a partnership and teams.

Extending the Enterprise



Evolution of IT System Architecture

► Centralized Computing





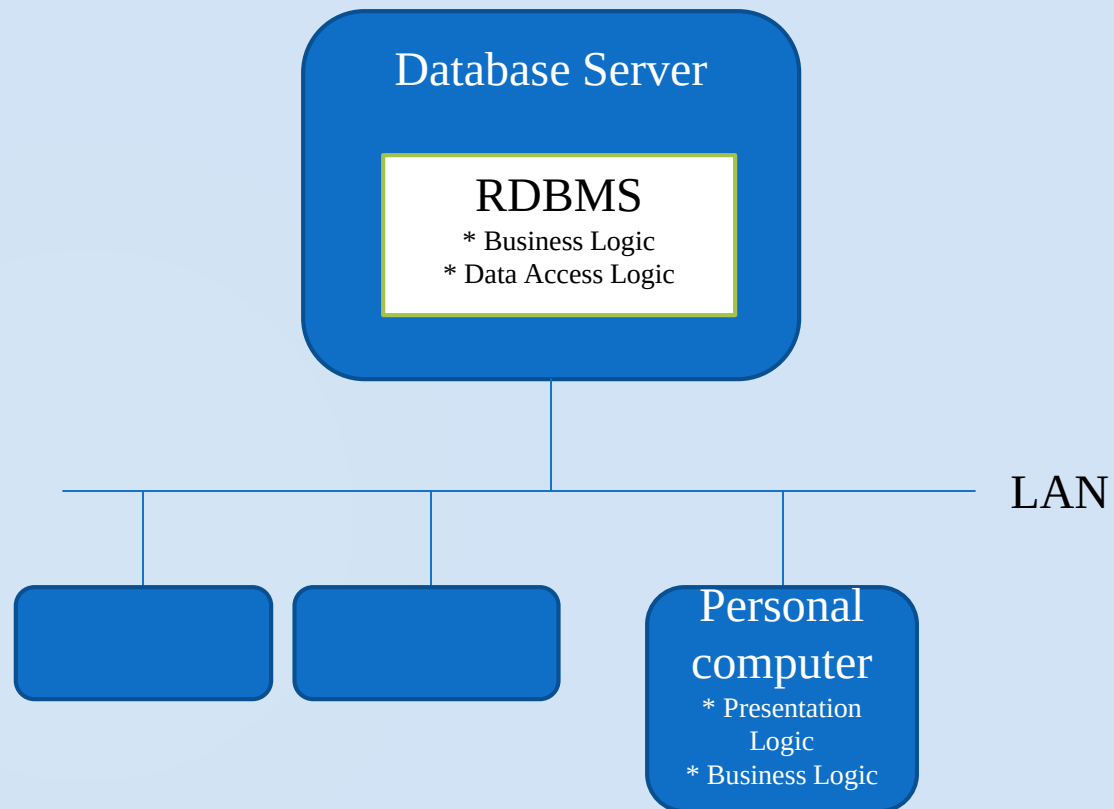
PC Networking: Moving Towards Decentralization

- ▶ **Client/Server Computing**

- ▶ Client/Server solutions involve the following independent layers:
 - ▶ **Presentation Logic** – how the user interacts with application
 - ▶ **Business Logic** – Everyday functions of the business
 - ▶ **Data Storage** – Handles the storage and retrieval of data

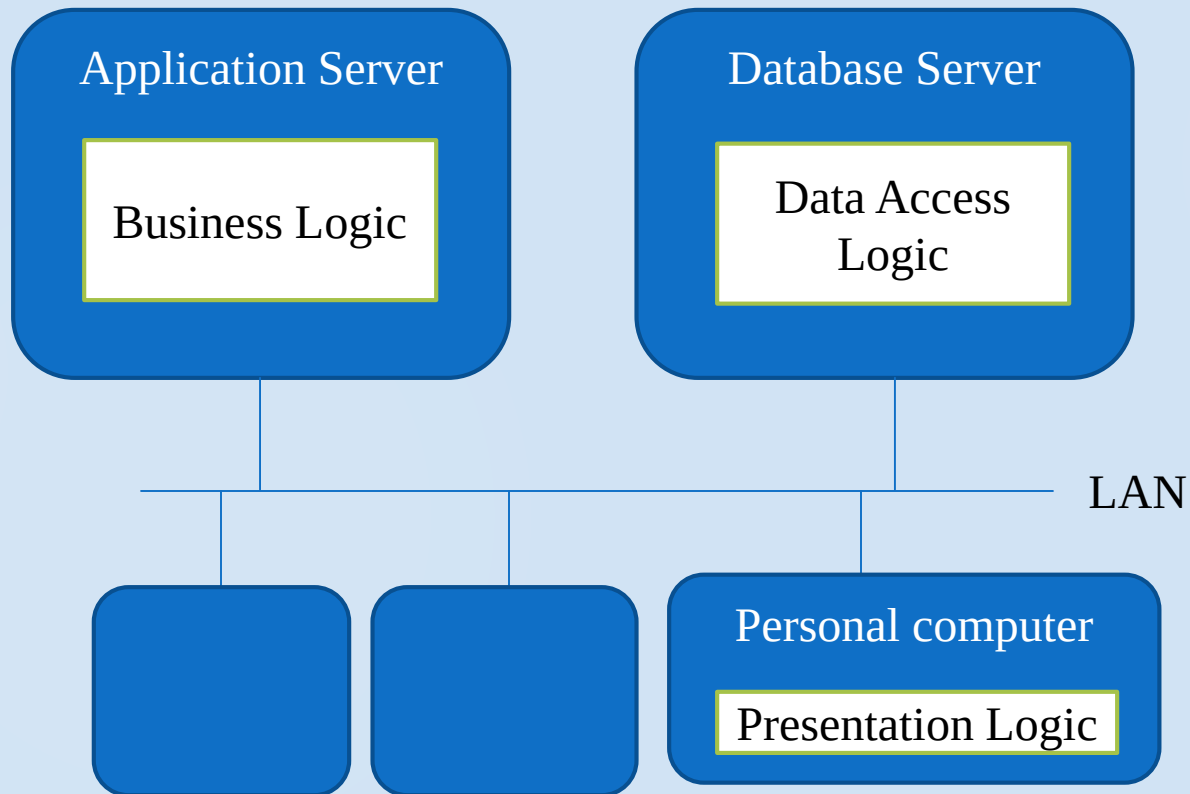
System Architecture

► Two Tier Client/Server



System Architecture

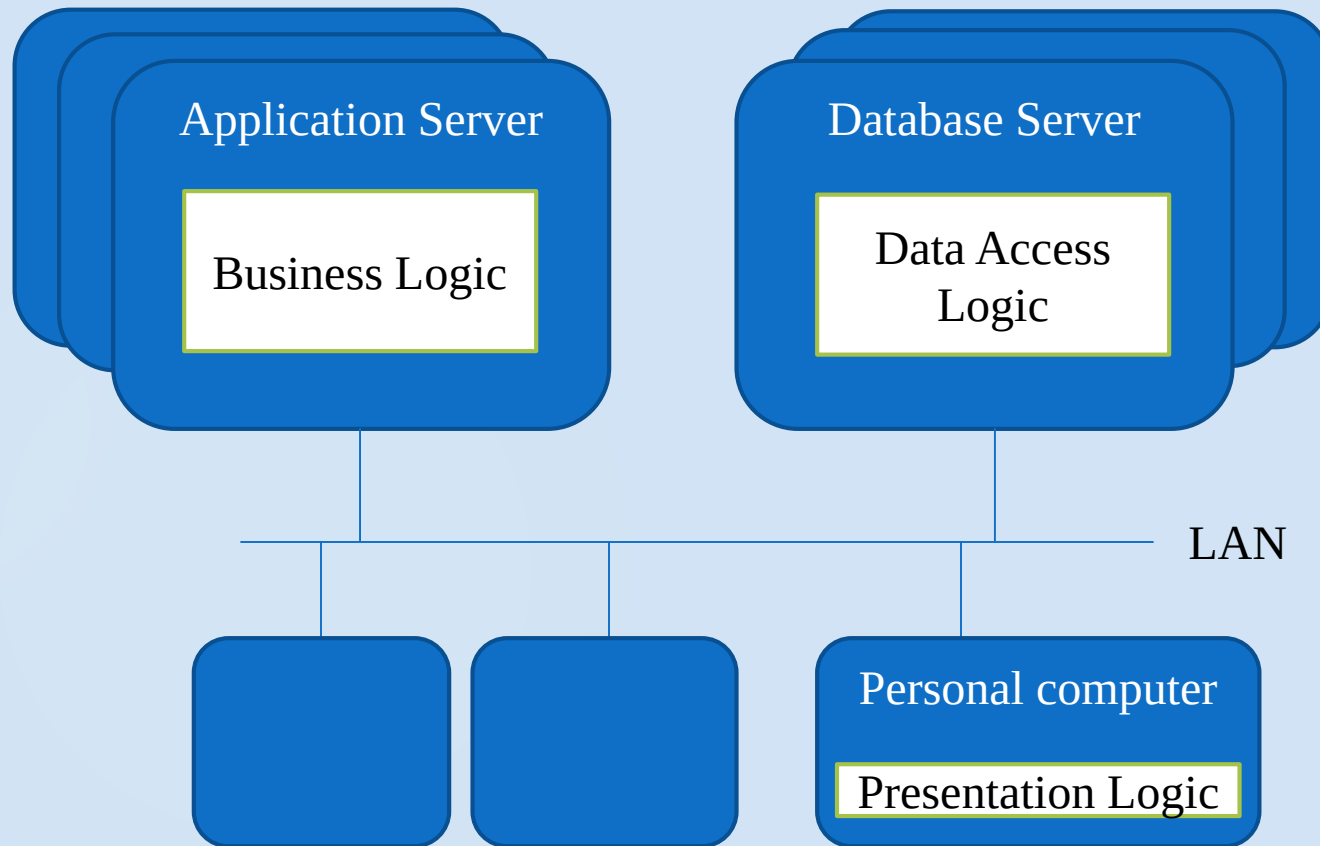
► Three Tier Client/Server



System Architecture



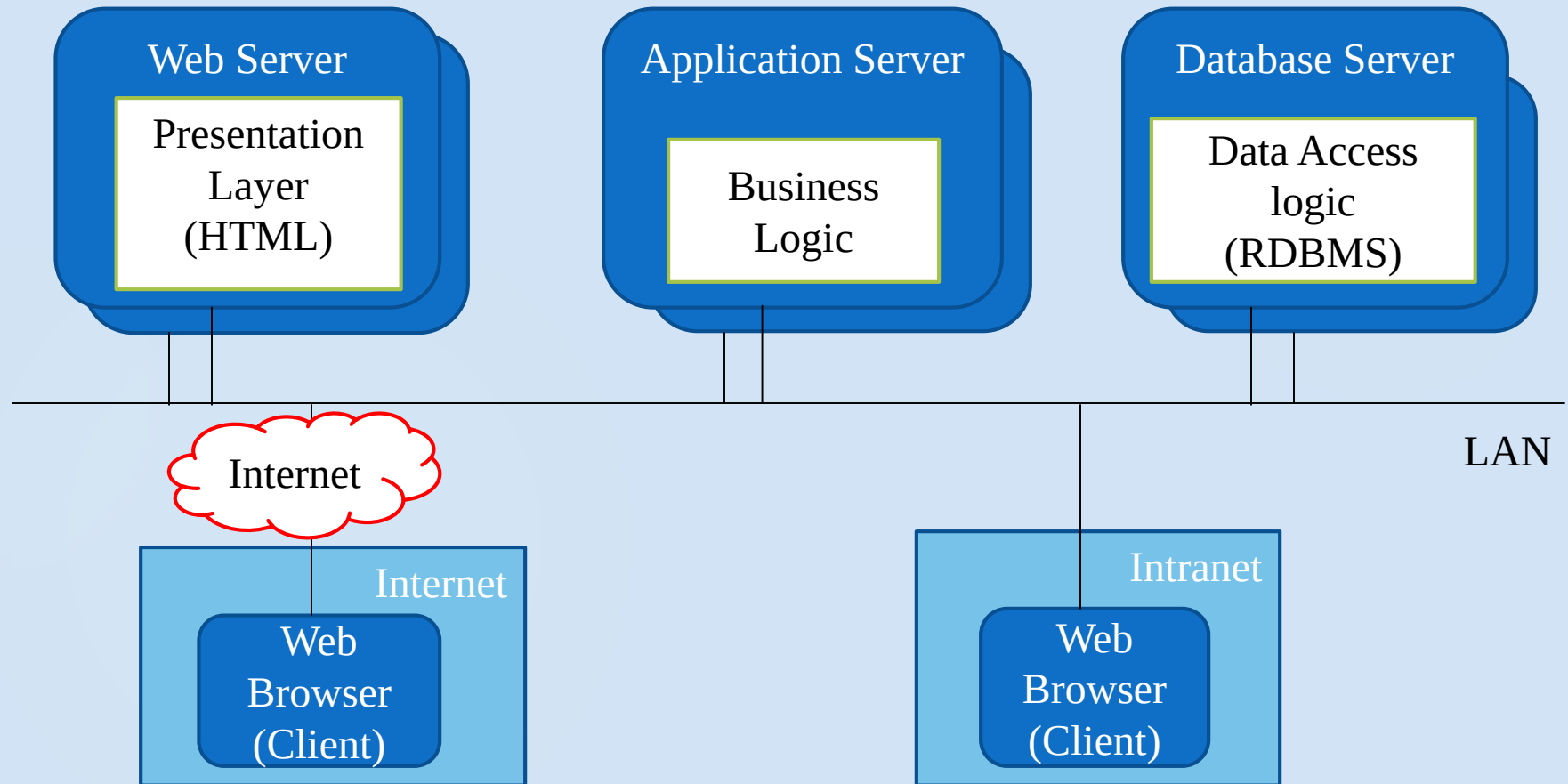
► Multi Tier Client/Server



System Architecture



► Web Technologies: Centralized Computing





Web Technology Fundamentals

- ▶ **ARPANET**

- ▶ *Advanced Research Projects Agency Network*

- ▶ **TCP/IP Networking**

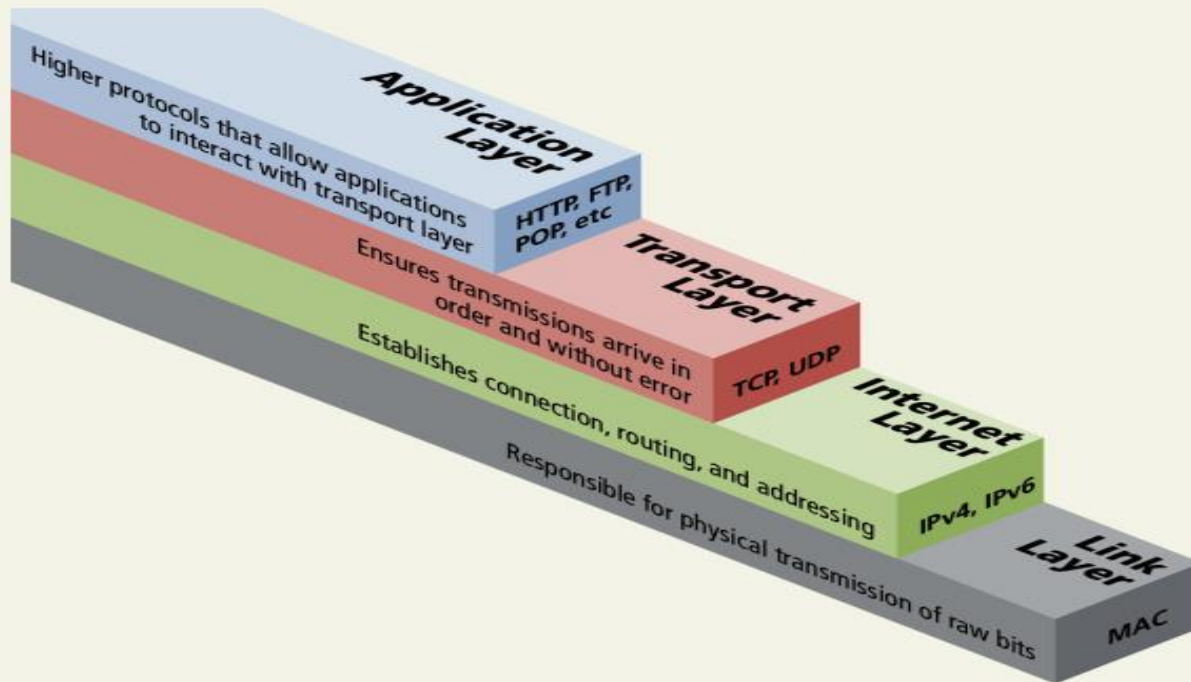
- ▶ **Internet Protocol** – It is used to route a packet of data from node to node across the network.
 - ▶ **Transmission Control Protocol** – It is used to create a point-to-point communication channel and ensures that the information is delivered error free and in correct order that it was originally transmitted.

Web Technology Fundamentals



► TCP/IP Protocol Stack

Four Layer Network Model



Web Technology Fundamentals



▶ Network Addresses

- ▶ Every computer that is connected to an *IP network* is uniquely identified by a value, known as its *IP address*.
 - ▶ Example: 194.27.254.189



Web Technology Fundamentals

- ▶ **Port Numbers** – It is used to distinguish between the individual networking applications that are running simultaneously above the TCP/IP protocol stack.
 - ▶ Port Numbers for standard TCP/IP services examples:
 - ▶ 80 – Hypertext Transfer Protocol (HTTP) ...Web/WWW
 - ▶ 21 - File Transfer Protocol (FTP) ... File Transfer
- ▶ **Uniform Resource Locator (URL)** – All information over the internet can be accessed, using a reference called URL.



Web Technology Fundamentals

► Uniform Resource Locator

► `<protocol> : // <machine id> / <local name>`

- ❖ `<protocol>` : identifies the application protocol used to retrieve the source.
- ❖ `<machine id>` : identifies the server on which the resource is located.
- ❖ `<local name>` identifies the resource, and that can include a directory structure.

Web Technology Fundamental



- **The World Wide Web (WWW)**
- Web is not a network but an application that operates over network using TCP/IP protocols.
- WWW is a service used world over for information sharing
- Introduced in 1990 by Tim Berners Lee
- One of the services available over Internet



Tim Berners Lee

Web Technology Fundamental (WWW service over internet)

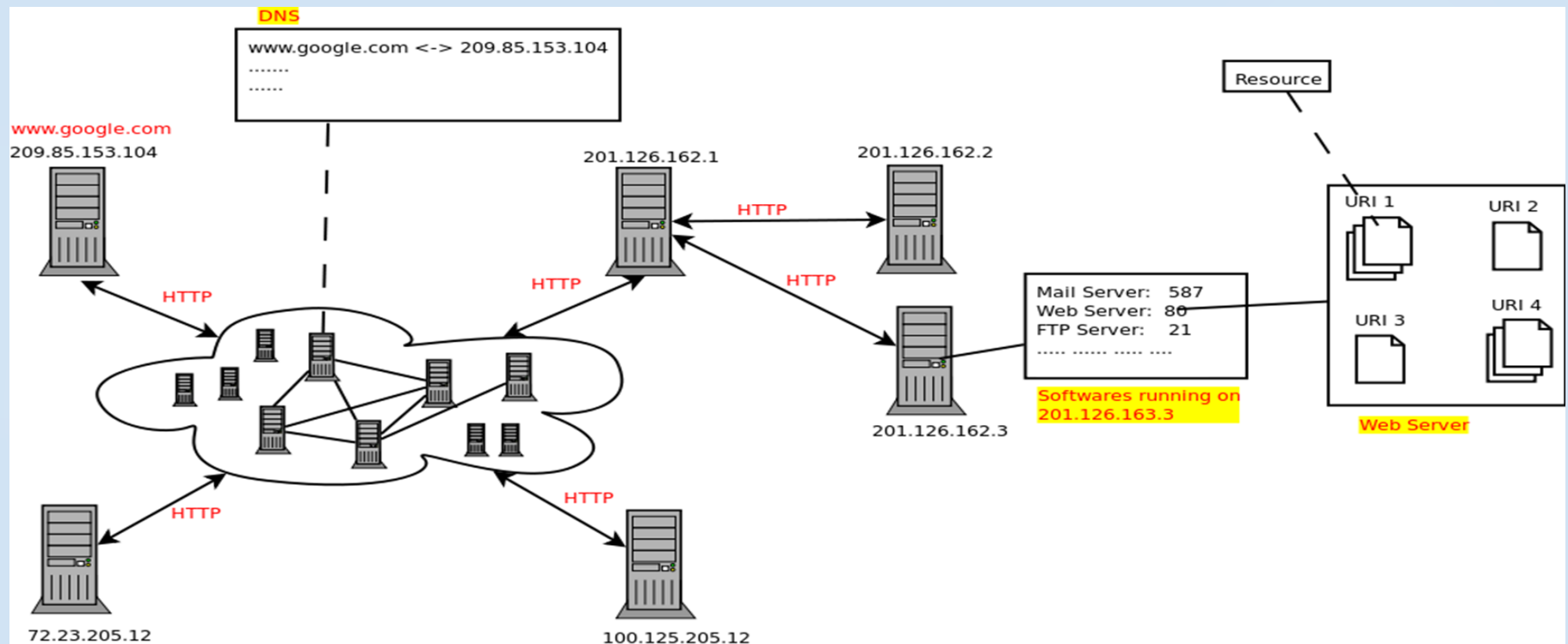


- Comprises of a client-server architecture
- Three major components
 - Web Server
 - Agent / Browser
 - HTTP Protocol
- Early contributing organization
 - CERN
- Current contributing organization
 - W3C

Web Technology Fundamental (WWW as Service internet)



- ▶ **Hyper Text Transfer Protocol (HTTP)** – protocol used to get the required document using HTTP request and response message.



Web Technology Fundamentals



- ▶ **Hypertext Markup Language (HTML)** – Page description language for defining the content of hypertext documents displayed in a format established by the Web browser rendering scheme.
 - ▶ HTML files are typically suffixed with either .htm or .html.

<HTML>

<HEAD>

<TITLE>

Document title

</TITLE>

</HEAD>

<BODY>

Document content

</BODY>

</HTML>

Internet



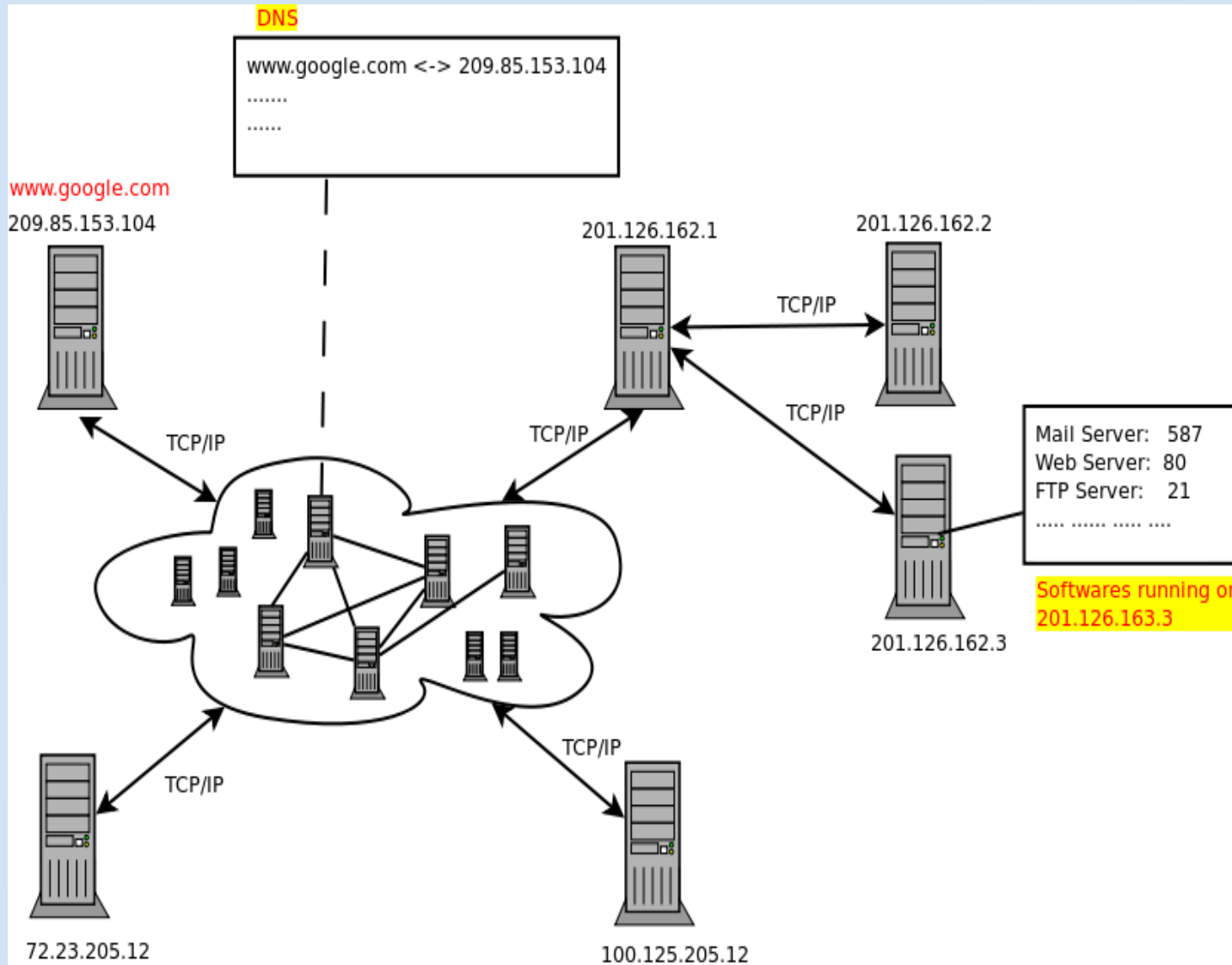
- **Internet** is a global network of computers communicating through **internet** over diverse underlying infrastructure
- The core enabler of internet is TCP/IP protocol suite worked out by Robert Kahn and Vint Cerf in 1973
- Early contributing organizations:
 - DARPA,
 - Stanford Research Institute
 - CERN
- Current organizations:
 - IETF
 - ICANN



Robert Kahn



Vint Cerf



Browser

- Browser function is to:
 - Fetch web content
 - Present web content
 - Efficient rendering
 - Secure communication
 - Support extensibility
- Early browsers
 - WorldWideWeb
 - Mosaic
 - Netscape Navigator
- Modern browsers
 - Google Chrome
 - Microsoft Internet Explorer



Mark Andreessen

1. What is the World Wide Web (WWW)?

The World Wide Web is a system of interlinked web pages and resources (like text, images, videos) that can be accessed through the internet using

🔗 Example: When you open google.com or wikipedia.org, you're using the WWW

2. What is the Internet?

The Internet is a global network of computers and servers connected together that allows communication and data exchange.

🔗 Example: The Internet makes email, chatting, online gaming, and the WWW possible.

3. Are WWW and Internet the same or different?

They are different but related:

The Internet = the big network (infrastructure).

The WWW = one of the services that runs on top of the internet (like email or WhatsApp).

🔗 So, WWW is a part of the Internet, not the same as it.

4. How the Web Works?

You type a web address (URL) in your browser.

Browser asks a DNS server to find the IP address of the website.

The request goes to the web server (where the site is hosted).

The server sends back the HTML, CSS, JS, images, etc.

Your browser displays it as a web page.

5. What it takes to program the web?

To make websites or web apps, a developer needs:

Frontend (Client-side): HTML, CSS, JavaScript (for design & interactivity).

Backend (Server-side): Languages like PHP, Python, Java, Node.js, C#.

Database: MySQL, MongoDB, PostgreSQL (for storing data).

Web server & hosting knowledge.

6. What issues warrant a web developer's attention?

Web developers need to focus on:

Security (protecting user data).

Performance (fast loading).

Responsiveness (works on mobile, tablet, PC).

Accessibility (usable by everyone, even disabled people).

Cross-browser compatibility (works on Chrome, Firefox, Edge, Safari).

Scalability (can handle more users if needed).

7. Difference between Web App and Desktop App

Feature Web App Desktop App

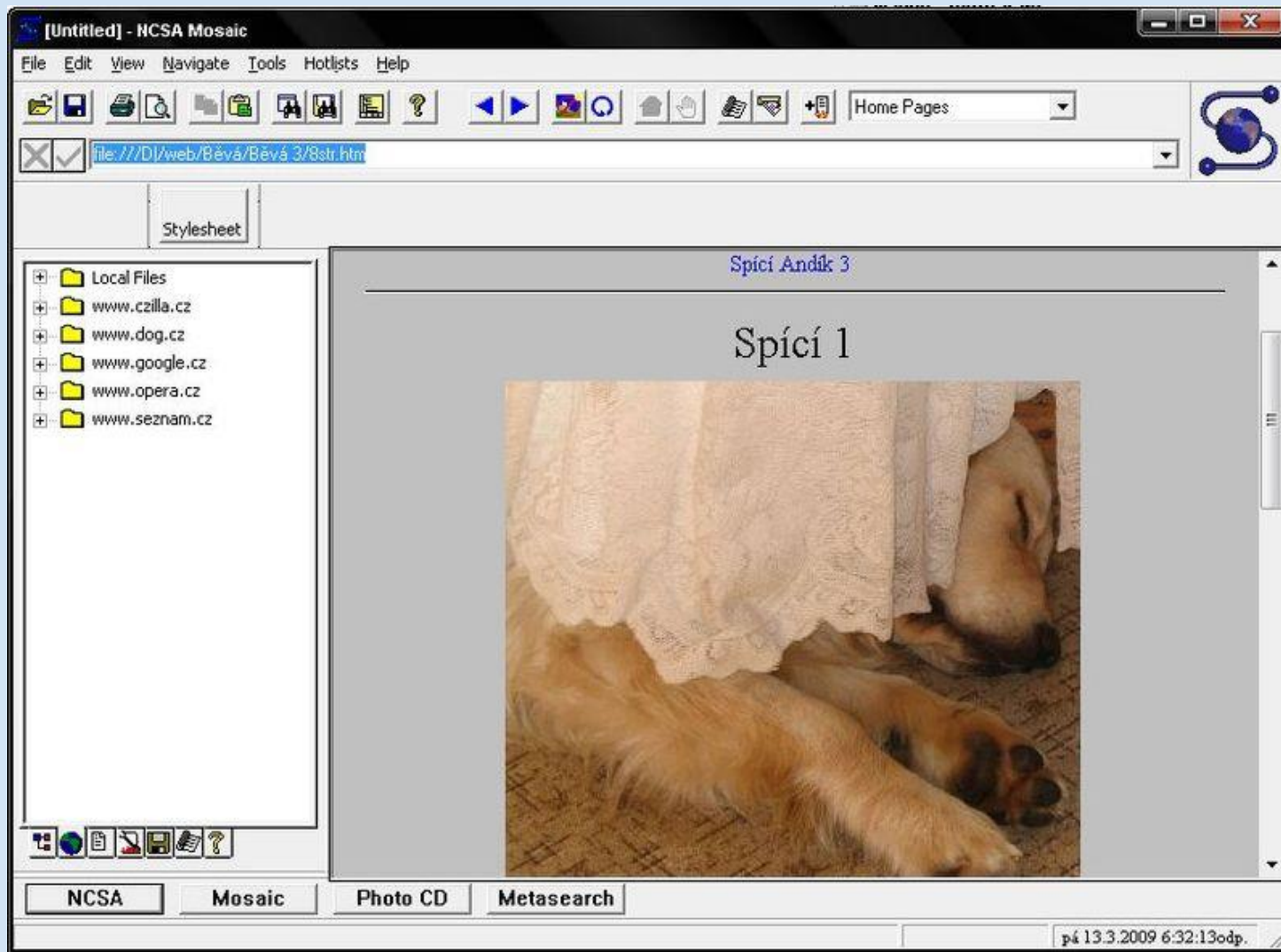
Runs on Browser (needs internet) Installed on PC

Example Gmail, Facebook MS Word, Photoshop

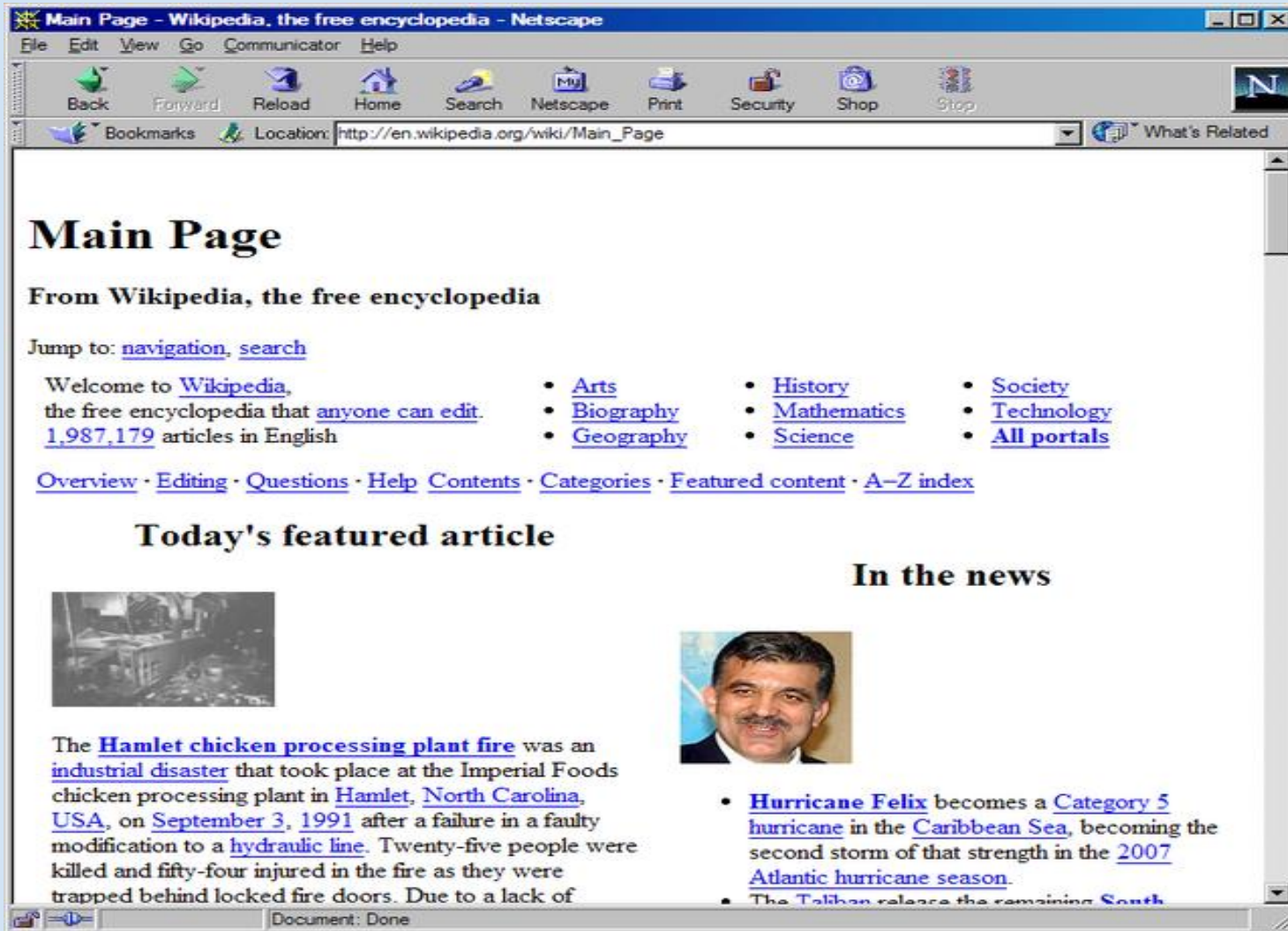
Access Any device with browser Only on the installed device

Update Automatic (on server) User must install updates

Storage Data stored on cloud/server Data stored locally



Mosaic browser by Mark Andreessen developed at UIUC (source : Wikipedia)



Netscape Navigator by Netscape Communications (source : Wikipedia)



Questions

- What is world-wide web?
- What is Internet?
- Are www and Internet same or different?
- How the web works?
- What it takes to program web?
- ***What issues warrant a web developers attention?***
- ***What difference between Web App and Desktop App ?***



Questions - What issues warrant a web developers attention?

- Understand the web architecture
- Understand technologies (client and server side both) that support efficient creation and consumption of web content for quality user experience
- Build architectures for integration of web-based applications with other aspects of underlying business for supporting diverse functions and allow extensibility in case of changing business needs
- Address performance and scalability issues
- Ensure security of communication over web